Arquitectura de Robots

INTRODUCTION

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Section 1.1

Robot

Robot

1 Robot

The term **robot** was first introduced by the Czech playwright **Karel Capek** in his **1920** play Rossum' s Universal Robots, the word robota being the Czech word for worker.

Since then, the term has been applied to a great variety of mechanical devices, such as teleoperators, underwater vehicles, autonomous cars, drones, etc.

Section 1.2

Industrial Manipulators

Industrial Manipulators

1 Robot

An **industrial manipulator** of the type shown in Figure 1 is essentially a mechanical arm operating under computer control.

It is nevertheless extremely complex electromechanical systems whose analytical description requires advanced methods, presenting many challenging and interesting research problems.

Figura 1: *A six-axis industrial manipulator.*



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2 Definition

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Robot Institute of America (RIA)

An official definition of such a robot comes from the **Robot Institute of America (RIA)**:

A robot is a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.

Advantages of the robots

2 Definition

Among the advantages often cited in favor of the introduction of robots are decreased labor costs, increased precision and productivity, increased flexibility compared with specialized machines, and more humane working conditions as dull, repetitive, or hazardous jobs are performed by robots.

Industrial manipulator

2 Definition

The industrial manipulator was born out of the marriage of two earlier technologies: teleoperators and numerically controlled milling machines.

Teleoperators, or master-slave devices, were developed during the second world war to handle radioactive materials.

Computer numerical control (CNC) was developed because of the high precision required in the machining of certain items, such as components of high performance aircraft.

First industrial robots

2 Definition

The first industrial robots essentially combined the mechanical linkages of the teleoperator with the autonomy and programmability of CNC machines.

The first successful applications of robot manipulators generally involved some sort of material transfer.

These first robots could be programmed to execute a sequence of movements, such as moving to a location A, closing a gripper, moving to a location B, etc., but had no external sensor capability.

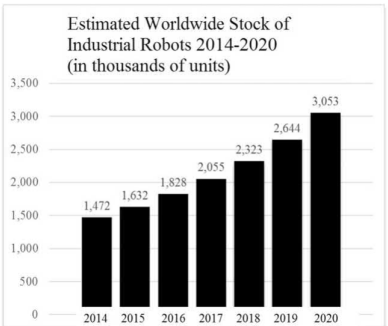
Industrial robots worldwide 2014–2020

2 Definition

Estimated number of industrial robots worldwide 2014–2020. The industrial robot market has been growing around 14 % per year. (Source: International Federation of Robotics 2018.)

In 2018 alone, more than 30 million service robots were sold worldwide.

Figura 2: *Industrial robots worldwide 2014–2020.*



Mobile Robot

2 Definition

Mobile robots encompass wheel and track driven robots, walking robots, climbing, swimming, crawling and flying robots. A typical wheeled mobile robot is shown in Figure 3.

Mobile robots are used as household robots for vacuum cleaning and lawn mowing robots, environmental monitoring, forestry and agriculture, and other applications.

Figura 3: *The Fetch series.*



Mechatronics

2 Definition

Mechatronics has been defined as the synergistic integration of mechanics, electronics, computer science, and control, and includes not only robotics, but many other areas such as automotive control systems.

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Section 3.1

Symbolic Representation of Robot Manipulators

Symbolic Representation of Robot Manipulators

Robot manipulators are composed of **links** connected by **joints** to form a **kinematic chain**.

Joints are typically rotary (**revolute**) or linear (**prismatic**).

Section 3.2

Types of joints

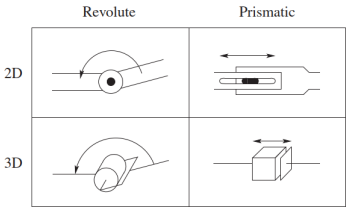
Types of joints

A revolute joint is like a hinge and allows relative rotation between two links.

A prismatic joint allows a linear relative motion between two links.

We denote revolute joints by R and prismatic joints by P, and draw them as shown in Figure 5.

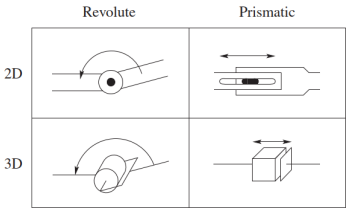
Figura 4: *Symbolic representation of robot joints.*



Type of joints symbolic

The **revolute joint** (shown in 2D and 3D on the left) **produces a relative rotation** between adjacent links.
The prismatic joint (shown in 2D and 3D on the right) **produces a linear** or telescoping motion between adjacent links.
For example, a three-link arm with three revolute joints will be referred to as an RRR arm.

Figura 5: *Symbolic representation of robot joints.*



Joints

Each **joint** represents the interconnection between two links.

We denote the axis of rotation of a revolute joint, or the axis along which a prismatic joint translates by z_i if the joint is the interconnection of links i and $i+1$.

The **joint variables**, denoted by θ for a revolute joint and d for the prismatic joint, represent the relative displacement between adjacent links

Section 3.3

The Configuration Space

The Configuration Space

A **configuration** of a manipulator is a complete specification of the location of every point on the manipulator.

The set of all configurations is called the **configuration space**.

vector q

We will represent a configuration by a set of values for the joint variables.

We will denote this vector of values by q, and say that the robot is in configuration q when the joint variables take on the values $q_1, \ldots, q_n$, with $q_i = \theta_i$ for a revolute joint and $q_i = d_i$ for a prismatic joint.

Section 3.4

Degrees of freedom

Degrees of freedom

An object is said to have n **degrees of freedom** (DOF) if its configuration can be minimally specified by n parameters.

The number of DOF is equal to the dimension of the configuration space. For a robot manipulator, the number of joints determines the number of DOF.

Position and orientation

A rigid object in three-dimensional space has six DOF: three for **positioning** and three for **orientation**.

Therefore, a manipulator should typically possess at least six independent DOF.

With fewer than six DOF the arm cannot reach every point in its work space with arbitrary orientation.

Redundante

Certain applications such as reaching around or behind obstacles may require more than six DOF. A manipulator having more than six DOF is referred to as a **Kinematically redundant** manipulator.

The State Space

A configuration provides an instantaneous description of the geometry of a manipulator, but says nothing about its dynamic response.

The **state** of the manipulator is a set of variables that, together with a description of the manipulator' s dynamics and future inputs, is sufficient to determine the future time response of the manipulator.

The **state space** is the set of all possible states.

The Workspace

The **workspace** of a manipulator is the total volume swept out by the end effector as the manipulator executes all possible motions.

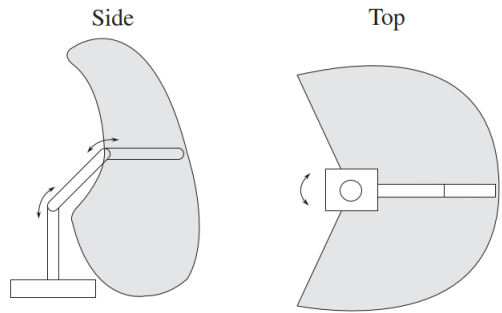
The workspace is constrained by the geometry of the manipulator as well as mechanical constraints on the joints.

For example, a revolute joint may be limited to less than a full 360° of motion. The workspace is often broken down into a **reachable** workspace and a **dexterous** workspace.

Reachable and dexterous

The **reachable** workspace is the entire set of points reachable by the manipulator, whereas the **dexterous** workspace consists of those points that the manipulator can reach with an arbitrary orientation of the end effector.

Figura 6: *Workspace Angular.*



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Q&A

Thank you for your attention

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